

MEDICAL PHYSICS–LED STRATIFIED DOSE OPTIMIZATION BENCHMARKS FOR DIGITAL BREAST TOMOSYNTHESIS

Supporting multidisciplinary breast care



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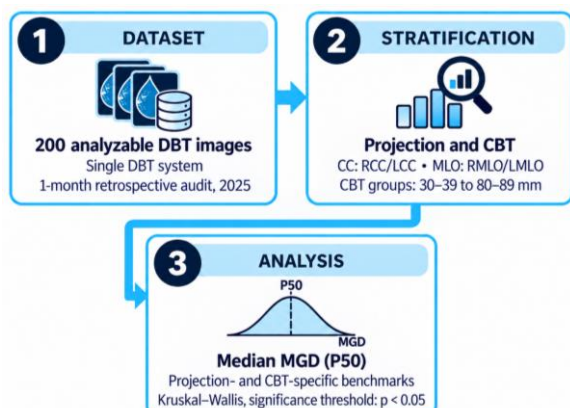
BACKGROUND

Dose optimization in DBT must balance **radiation dose and diagnostic image quality**. Conventional single-value institutional benchmarks may mask dose variation across **projection type** - craniocaudal (CC) and mediolateral oblique (MLO) - **and compressed breast thickness (CBT)**. **Stratified benchmarks** may therefore provide a more clinically meaningful approach to dose optimization.

OBJECTIVE

To establish **projection- and CBT-specific mean glandular dose (MGD) benchmarks** for DBT using the **50th percentile**, supporting more individualized dose optimization.

METHODS



RESULTS | STRATIFIED BENCHMARKS

MGD (P50) by Projection and CBT

View	30–39	40–49	50–59	60–69	70–79	80–89
RCC	1.08	1.54	1.92	2.41	2.94	—
LCC	1.21	1.49	1.86	2.41	2.97	—
RMLO	1.19	1.50	1.74	2.41	3.08	3.46
LMLO	—	1.50	1.92	2.37	3.08	3.70

Compressed breast thickness (mm)

MGD increased progressively with CBT, with significant differences across CBT categories for all projections (Kruskal–Wallis, all $p < 0.001$).

MGD ↑ WITH CBT

1.08 → 3.70 mGy

MGD increased progressively across CBT categories.

MLO > CC AT
HIGHER CBT

3.70 vs 2.97 mGv

Higher maximum MGD was observed in thicker MLO projections.

WHY STRATIFICATION MATTERS

Greater between-view variation at higher CBT

Supports projection- and CBT-specific benchmarking.

CONCLUSION / TAKE-HOME MESSAGE

1. MAIN FINDING

Projection- and CBT-based stratification reveals dose variation hidden by a single benchmark.

2. CLINICAL MEANING

More individualized benchmarks can guide targeted optimization while preserving image quality.

3. NEXT STEP

Embed stratified MGD review into routine medical physics QA and continuous quality improvement.

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